

Abstract of Scientific Justification (300 words max) (2 points)

Supernovae are powerful stellar explosions that happen near the end of some stars' lives. However, not all supernovae form in the same way. Core-collapse supernovae are linked to the deaths of massive stars (Smartt 2009), while Type Ia supernovae come from the thermonuclear explosion of a white dwarf (Maoz et al. 2014). Studying these different events gives us a way to see how stars can end their lives and what happens as they explode.

One of the main ways we can study a supernova is by following its brightness over time. With observations from multiple nights, we can put these measurements together to make a light curve and see how quickly the supernova brightens or fades. Taking images through different filters also lets us follow changes in color (de Jaeger et al. 2019). Depending on when we observe the supernova and how much of its light curve we are able to cover, these measurements can also help us compare it with known supernovae and possibly learn more about its type.

For this project, we plan to observe one or two recently discovered supernovae with the Mont4K CCD on the 61-inch Kuiper Telescope. We will mainly use the B and V filters, along with R if possible. We will measure each supernova relative to nearby comparison stars and use those measurements to build multi-band light curves. Since these objects can change quite a bit over a short period of time, choosing our final targets too early would not be practical. Instead, we will check the Transient Name Server shortly before our observing night and look for targets with $V < 19$ and declinations between approximately -5° and $+40^\circ$, and also check if their separation from the moon is large enough for us to get good results.

Telescope Requested (0.5 points): 61" Kuiper Telescope

Instrument (0.5 points): Mont4K CCD, using primarily the B, V, and if permitted, R filters.

Tolerable Moon Brightness (dark, grey, bright, etc) (0.5 points):

Dark or grey time would be preferred since our targets are relatively faint, around 18–19 mag. Bright time could still be used if the supernova is bright enough and sufficiently far from the Moon. However, moonlight increases the sky background, which can lower the signal-to-noise ratio and make observations of fainter supernovae more difficult.

Target List (2 points)

List potential targets you may observe and their basic properties like brightness. Under visibility write what months/days the target is up. For example "Object can be observed Feb-March."

Object Name	Right Ascension (h, m, s)	Declination (d, m, s)	V magnitude*	Object Type	Visibility
AT 2026aawd	20:21:37.093	+23:41:09.61	18.01	Transient (SN candidate)	September, October
AT 2026aars	02:29:34.883	+23:06:58.44	18.42	Transient (SN candidate)	September, October

* These are the discovery magnitudes, as stated on the Transient Name Server.

Since supernovae can change quite a bit in a short amount of time, any target we choose now may be at a very different stage by the time of our observing night. It could become too faint to observe or may have already passed its rise to peak brightness. Because of this, we will keep our target selection flexible. If a newly discovered supernova better meets our magnitude, declination, altitude, and Moon-separation requirements, we can use it instead of one of our current targets (which is most likely going to be the case).

Experimental Design & Technical Description (4.5 points total)

How much time is required for your program and what signal-to-noise is necessary for your observations? (2 points)

We are aiming for a signal-to-noise ratio of about 100 for each B, V, and R measurement so that we can get reliable brightness measurements. Using the simple approximation

$$\text{SNR} \approx \sqrt{Ft}$$

the exposure time needed for an SNR of 100 can be written as

$$t \approx 10,000 / F$$

where F is the count rate and t is the exposure time.

To get a rough idea of what our exposure time might be, we can use previous observations taken with the same Kuiper Telescope and Mont4K instrument. Xu et al. (2018) observed a source with a magnitude of about 16.2 and reached an SNR of around 300 using 30-second exposures. From this, we can estimate the count rate:

$$F_1 = 300^2 / 30 \approx 3000 \text{ counts/s}$$

Our faintest targets will be around V = 19, which is about 2.8 magnitudes fainter. We can account for this difference using

$$F_2 / F_1 = 10^{-0.4(m_2 - m_1)}$$

This gives

$$F_2 = 3000 \times 10^{-0.4(19 - 16.2)} \approx 228 \text{ counts/s}$$

Using this count rate, the exposure time needed to reach an SNR of 100 would be

$$t = 10,000 / 228 \approx 44 \text{ seconds}$$

Based on this, we can use about 45 seconds as a starting estimate for our exposure time. The actual exposure will probably be different depending on the filter, target, and conditions that night. Xu et al. (2018) also used a broad white-light filter rather than the B, V, and R filters we plan to use, so this is only meant to give us a rough starting point. We can take a few short test

exposures during the observing night and adjust the exposure time from there and account for actual observing conditions.

How will these observations contribute toward the accomplishment of the goals outlined in the abstract? **(2 points)**

Our observations will give us brightness measurements of each supernova in three optical bands. By comparing the measurements between different bands, we can see how the supernova's color changes over time, along with changes in its brightness. We can then combine our data with observations from other observing nights or with existing archival data to build a more complete light curve. Having more observations will give us a better picture of the shape of the light curve and allow us to measure how quickly the supernova rises or fades. With enough data, the light curve may also help us determine the type of supernova.

Back Up Plan. Describe a potential project that can be done with poor weather or archival data. **(0.5 points)**

We might use archival data in this case. I was thinking of making a light curve of maybe one or two active galaxies using archival NEOWISE data. We already have measurements in the W1 and W2 infrared bands taken over several years, so we could plot their brightness over time and look for any changes. We could also compare the W1 and W2 measurements to see if the color of the source changes along with its brightness. This would give us a simple way to study the long-term variability of an active galaxy if we are unable to collect our own observations because of bad weather.

References

de Jaeger, T., Zheng, W., Stahl, B. E., et al. 2019, MNRAS, 490, 2799–2821

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Maoz, D., Mannucci, F., & Nelemans, G. 2014, ARA&A, 52, 107–170

<https://doi.org/10.1146/annurev-astro-082812-141031>

Smartt, S. J. 2009, ARA&A, 47, 63–106

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Xu, S., Su, K. Y. L., Rogers, L. K., et al. 2018, ApJ, 866, 108

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